MEDINOVA TEACHING HOSPITAL

Class 06 — Cardiology

Heart Failure · Arrhythmias · ACS · Valvular Disease

> *"The heart speaks in rhythms.*
> *Interpret every beat."*

CLERKSHIP SERIES | MODULE 06 of 20

SECTION 01

ECG Systematic Interpretation Framework

A structured 5-step approach to avoid diagnostic errors in Electrocardiography (ECG).

Component	Assessment Step	Normal Reference Values
1. Rate & Rhythm	Calculate via 300/large squares. Assess P-waves.	Rate: 60-100 bpm. Rhythm: Sinus (P before every QRS).
2. Axis	Look at Leads I and aVF.	Normal: -30° to +90° (Both I and aVF are positive).
3. Intervals	PR interval, QRS duration, QTc.	PR: 120-200ms | QRS: < 120ms | QTc: < 440ms (M) / 460ms (F).
4. Morphology	ST-segment, T-waves, Pathological Q-waves.	ST: Isoelectric. T: Upright (except aVR/V1). Q: < 0.04s.
5. Hypertrophy	Sokolow-Lyon Criteria (V1 S + V5/V6 R).	LVH if sum > 35mm.

Interval Meanings:

- **PR Interval:** Time for impulse to travel from Atria to Ventricles.
- **QTc (Corrected QT):** Measures ventricular repolarization; prolonged QT risks *Torsades de Pointes* (fatal rhythm).

Axis Deviations:

- **Left (LAD):** Lead I (+), aVF (-). Suggests LVH or LBBB.
- **Right (RAD):** Lead I (-), aVF (+). Suggests RVH or PE.

Prof. Obi's Pearl #1 "The ECG is a snapshot in time. A normal ECG in a patient with active chest pain is not a 'clear' signal—it's a reason to repeat the test in 15 minutes."

SECTION 02

Chronic & Acute Heart Failure

HFrEF vs HFpEF Classification

Category	Ejection Fraction (EF)	Pathophysiology
HFrEF (Reduced)	≤ 40%	Systolic failure; the heart cannot pump effectively.
HFmrEF (Mildly Reduced)	41% - 49%	Transitional zone; managed similar to HFrEF.
HFpEF (Preserved)	≥ 50%	Diastolic failure; the heart cannot fill properly (Stiff heart).

The Four Pillars of GDMT (Guideline-Directed Medical Therapy)

Every HFrEF patient requires these four classes to reduce mortality:

1. **ARNi / ACEi:** (e.g., Sacubitril/Valsartan). Reduces afterload and remodelling.
2. **Beta-Blocker:** (e.g., Bisoprolol). Slows HR, reduces oxygen demand.
3. **MRA:** (e.g., Spironolactone). Blocks aldosterone-mediated fibrosis.
4. **SGLT2i:** (e.g., Dapagliflozin). Improves diuresis and metabolic efficiency.

Acute Management: The "Wet & Cold" Framework

- **Warm & Wet (90%):** Congested but perfused. Treatment: High-dose Loop Diuretics (Furosemide).
- **Cold & Wet:** Congested and Hypoperfused (Shock). Treatment: Inotropes (Dobutamine) + Urgent Specialist help.

Prof. Obi's Pearl #2 "Fluid in the lungs is a symptom; a failing pump is the cause. Don't just dry them out; fix the neurohormonal drive that's killing the myocardium."

SECTION 03

Acute Coronary Syndromes (ACS)

Syndrome	ECG Findings	Troponin	Pathology
Unstable Angina	Normal or ST-depression	Negative	Incomplete occlusion.
NSTEMI	ST-depression / T-wave inversion	Positive	Sub-endocardial infarct.
STEMI	ST-elevation / New LBBB	Positive	Transmural (Full) occlusion.

SECTION 04

Common Arrhythmias & Management

Atrial Fibrillation (AF)

• **Management:** Rate control (Bisoprolol/Digoxin) vs Rhythm control (Flecainide/Amiodarone).
• **Stroke Prevention:** Use **CHA_2DS_2-VASc** score. If ≥ 2 in men or ≥ 3 in women, start DOAC (Apixaban).

Acute Tachycardia Algorithm

- **Unstable?** (Shock, Syncope, Myocardial Ischemia) → Immediate Synchronized DC Cardioversion.
- **Stable Narrow QRS:** Vagal maneuvers → Adenosine (6mg-12mg-12mg).
- **Stable Broad QRS:** Assume VT. Loading dose of Amiodarone (300mg over 1 hour).

Prof. Obi's Pearl #3 "Broad complex tachycardia is VT until proven otherwise in the morgue. Never assume it's SVT with aberrancy in an unstable patient."

SECTION 05

Valvular Heart Disease Comparison

Lesion	Murmur Character	Classic Sign / Symptom	Intervention
Aortic Stenosis	Ejection Systolic (Crescendo-Decrescendo)	SAD: Syncope, Angina, Dyspnoea. Narrow pulse pressure.	TAVI / SAVR if symptomatic.
Mitral Regurg	Pan-systolic (Radiates to Axilla)	Displaced Apex Beat. Pulmonary hypertension.	Repair > Replacement.
Aortic Regurg	Early Diastolic (Decrescendo)	Water-hammer pulse. Wide pulse pressure.	Valve replacement.
Mitral Stenosis	Mid-Diastolic (Low pitch rumble)	Malar Flush (pink cheeks). Opening snap.	Balloon Valvuloplasty.

SECTION 06

Hypertension (NICE NG136)

Step 1 Treatment:
• Under 55 & non-black: **A** (ACEi/ARB).
• Over 55 or black: **C** (Calcium Channel Blocker).

Step 2: A + C or A + D (Diuretic).

Emergency vs Urgency:
• **Emergency:** BP > 180/120 + End-organ damage (Stroke, Encephalopathy). Requires IV Labetalol.
• **Urgency:** BP > 180/120 but patient is asymptomatic.

SECTION 07

Infective Endocarditis (IE)

Diagnosis requires the **Modified Duke Criteria**: 2 Major, 1 Major + 3 Minor, or 5 Minor.

Major Criteria:
1. 2x Positive Blood Cultures (typical organism).
2. Echo evidence (Vegetation, abscess, new valve regurg).

Minor Criteria:
- Predisposition (IVDU/Valve).
- Fever > 38°C.
- Vascular signs (Janeway lesions).
- Immunological signs (Osler nodes).

Empirical Antibiotic Strategy:

- **Native Valve:** Amoxicillin + Gentamicin.
- **Prosthetic Valve:** Vancomycin + Gentamicin + Rifampicin.

Prof. Obi's Pearl #4 "If a patient has a new murmur and a fever, it is Endocarditis until three sets of blood cultures and a trans-oesophageal echo prove you wrong. Do not delay."

Medical Meanings:
- **Cardioversion:** Restoring normal rhythm via electrical shock.
- **Inotrope:** A drug that increases the force of heart muscle contraction.
- **Vegetation:** A clump of bacteria and debris attached to a heart valve.
- **TAVI:** Transcatheter Aortic Valve Insertion (minimally invasive).

OFFICIAL SIGN-OFF

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